

In both versions, the environment worked as expected. The agent received percepts such as breeze, stench, glitter, bump, and scream, then selected actions and received rewards from the environment. The random agent often repeated poor actions and eventually entered a dangerous square, which led to a large negative reward.

1. Using only the percepts for training can be a problem because they do not show the full state of the world. They only give local clues about nearby hazards or rewards. The same percept can appear in different situations, so the agent may not know which action is best. This makes the problem partially observable and harder to learn.
2. A table could be used for the value function if we include the full state, because WumpusWorld is small and has a limited number of positions, orientations, and actions. However, if the table only uses percepts, it may not work well because different states can produce the same percept. So a table is possible, but only if the state representation includes enough information.
3. Curriculum training means starting with an easier version of the environment and making it harder over time. For example, there could be fewer pits, no Wumpus, or the gold could be placed closer to the start. This would help the agent learn to explore and collect the gold instead of always leaving immediately. After the agent learns useful behavior, the normal harder environment can be introduced.
4. A problem is Markovian if the current state contains all the information needed to determine the next state and reward, given the current action. In other words, if a process is Markovian, then the future depends only on the present state and action, not on the full history of what happened before. In WumpusWorld, the answer depends on how the state is defined. If we use the full hidden environment state, then the problem is Markovian. This full state would include the agent's position and orientation, the locations of the pits, gold, and Wumpus, whether the Wumpus is alive, and whether the agent still has the arrow or the gold. Given all of this information and the chosen action, the next state and reward can be determined without needing earlier history. However, from the agent's point of view, WumpusWorld is not fully Markovian if the agent only receives percepts such as breeze, stench, glitter, bump, and scream. These percepts do not reveal the complete world state. For example, a breeze tells the agent that a pit is nearby, but it does not say exactly where the pit is. The same percept can occur in several different states, and the best action may depend on what the agent has observed in previous steps. This means the current percept alone is not enough to fully determine the next decision. Because of this, WumpusWorld is Markovian at the

level of the full environment state, but not fully Markovian from the limited perspective of the agent. From the agent's point of view, it is better described as a partially observable problem, or POMDP, rather than a fully observable MDP.